Assessment Tool - 2

Concept Mapping

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1) Machine Learning System :- It is a technique in which a computer learns patterns from data and make predictions or decisions without being explicitly programmed.

Data collection
↓
Data preprocessing
↓
Feature selection
↓
Model Training
↓
Model Evaluation
↓
Prediction.

Main Blocks:-

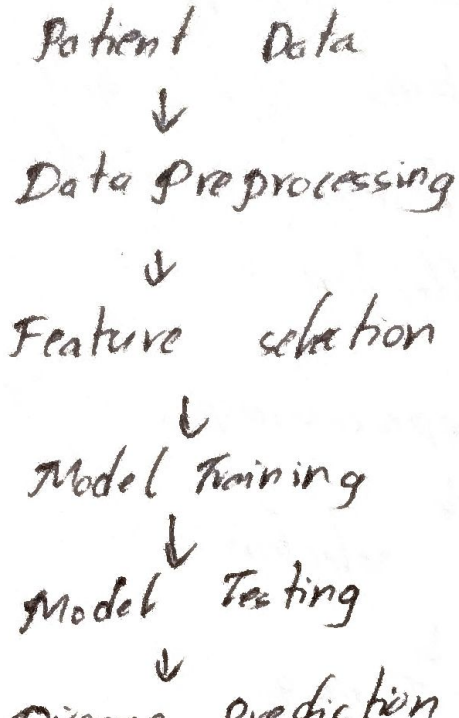
1) Data collection :- Collect data from different sources.
2) Preprocessing :- Remove errors, duplicate, data & handle missing.
3) Feature selection : select the important attributes required for prediction.
4) Model Training : Train the ML algorithm using training data
5) Evaluation : check the accuracy and performance of the model.
6) Prediction : Use the trained model to predict new data.

Types :- 1) Supervised Learning :- Uses labelled data Ex : Spam email detection.
2) Unsupervised learning :- Uses Unlabelled data. Ex : Customer clustering.

Conclusion:- ML systems learn from data and make useful predictions or identify hidden patterns.

(6) Health Analysis Using Machine Learning

A machine learning system can be used to predict diseases as heart disease and diabetes from patient data.



Important Features.

* Age
* Blood pressure
* Glucose level
* weight
* Lifestyle
* Exercise.

steps:

1) Collect Data:- Gather historical patient records
2) Preprocess data: Handle missing values & incorrect data.
3) select Features: Choose important health parameters.
4) Train model: Train the ML algorithm using previous patient data.
5) Evaluate Model:- Test the accuracy of the model.
6) Predict: Give new patient data to predict disease risk.

Algorithm:- Logistic Regression, DecisionTree & Random forest.

Ex:- If a patient has high glucose, high blood pressure and other risk factors, the model can predict a

higher disease risk.

Conclusion: The system can help in early disease - r... identification.

Machine Learning Pipeline:

A MLP is a sequence of stage used to develop and deploy a ML model.

Data collection
↓
Preprocessing
↓
Feature Engineering
↓
Model Training
↓
Evaluation
↓
Deployment.

Explanation:-

1) Data collection :- Gather required data.
2) Preprocessing :- clean the data and handle missing values.
3) Feature Engineering :- Select or create useful features.
4) Model Training :- Train the ML model using Training data.
5) Evaluation :- Measure the performance using test data.
6) Deployment :- use the trained model in a real application.

Errors :- Training Error :- Error obtained when the model is tested on training data.

Generalization Error :- Error obtained when the model is tested on new, unseen data.

Ex:- Training accuracy = 95%.
Testing accuracy = 85%.
the model should have good performance on bo
training and new data.

Conclusion:- A proper ML pipeline help develop a
accurate and reliable model.

12) Spam Email classification :- It is a supervised learn
application used to identify unwanted emails.

Email Dataset
↓
Tokenization
↓
stop-word Removal
↓
Vectorization
↓
ML Algorithm
↓
Spam / Not spam

Steps:-
1) Data collection:- Collect emails labelled as Spam or Not spam.
2) Tokenization :- Divide the email into individual words.
3) stop-word Removal :- Remove common words such as the, is, a.
4) Vectorization :- Convert text into numerical values.
5) Training :- Train the ML model using labelled emails.
6) classification :- classify a new email as spam or Not spam.

Algorithms:- * Naive Bayes
* Support Vector Machine (SVM)
* Logistic Regression

Evaluation:- The model can be evaluated using
* Accuracy * Recall
* Precision * F1-score

ML Model

↓

Spam.

Conclusion:- ML helps automatically detect Spam & improve email management.

3) Clustering and K-Means

Clustering:- is an unsupervised learning technique that group similar data objects into.

Ex:- customers can be grouped based on their purchasing behavior.

K-Means Algorithm:-

K-means divides the dataset into k clusters

steps:-

1) Choose k: select the required no. of clusters.
2) Initialize centroids:- select intial cluster centers.
3) Assign points: Assign each point to its nearest centroid.
4) Calculate centroids:- Find the mean of points in each cluster
5) Repeat: Continue until the centroids become stable.

Ex:- Given 10, 11, 12, 50, 51, 52

k=2 cluster 1 -> 10, 11, 12

cluster 2 -> 50, 51, 52

Elbow Method:-

The Elbow method is used to select the appropriate value of k. we calculate the error for different k values and identify the point where the reduction in error becomes small. this point is called

K-means	Hierarchical
Uses centroids	Uses hierarchy
Requires k	k is not initially required
Faster	Comparatively slower
Produces clusters	Produces tree-like structure.

Conclusion:- K-Means is a simple and commonly used algorithm for grouping similar data.